

Image Restoration

What is Image Restoration

- The purpose of image restoration is to restore a degraded/distorted image to its original content and quality.
- Restoration involves following process:-
- Modeling of Degradation
- Applying the inverse process to recover the original image

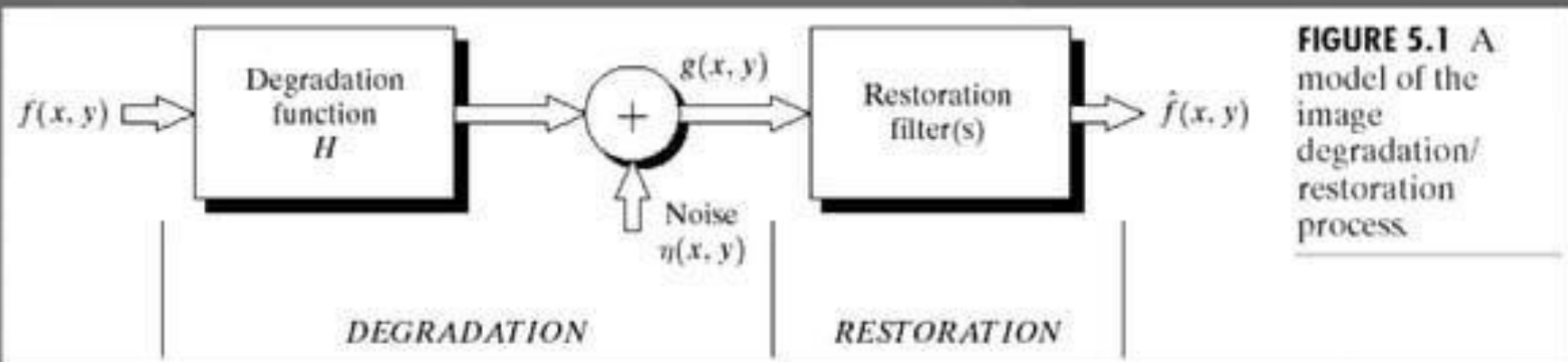


FIGURE 5.1 A model of the image degradation/restoration process.

Explanation of diagram

- We will assume that a degradation function exists, which, together with additive noise, operates on the input image $f(x,y)$ to produce a degraded image $g(x,y)$.
- The objective of restoration is to obtain an estimate for the original image from its degraded version $g(x,y)$ while having some knowledge about the degradation function H and the noise

$$\eta(x,y).$$

Model for image degradation/restoration process

- the degraded image in the **spatial domain** is

$$g(x, y) = h(x, y) \underset{\substack{\uparrow \\ \text{convolution}}}{\otimes} f(x, y) + \eta(x, y)$$

- Therefore, in the **frequency domain** it is

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

Degradation models : noise only

$$g(x, y) = f(x, y) + \eta(x, y)$$

$$G(u, v) = F(u, v) + N(u, v)$$

- ⊙ Some important noise probability density functions
 - ⊙ – Gaussian noise
 - ⊙ – Rayleigh noise
 - ⊙ -Erlang gamma noise
 - ⊙ – Exponential noise
 - ⊙ – Impulse

Gaussian Noise

- The PDF of a Gaussian random variable z given by:

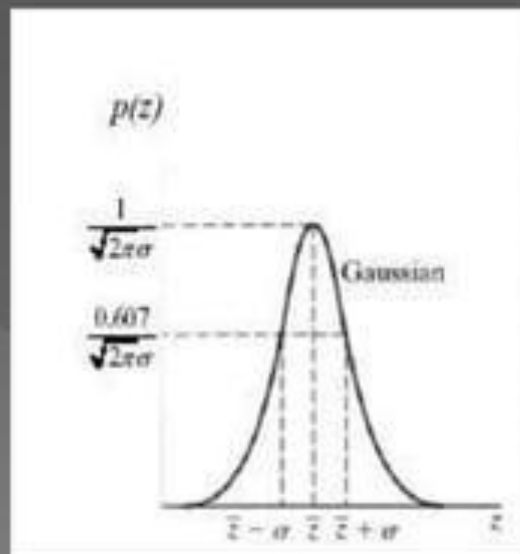
$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\bar{z})^2}{2\sigma^2}}$$

z represents intensity,

$\bar{z}$ is the mean (average) value of z ,

σ is its standard deviation.

σ^2 is the variance of z .



Rayleigh noise

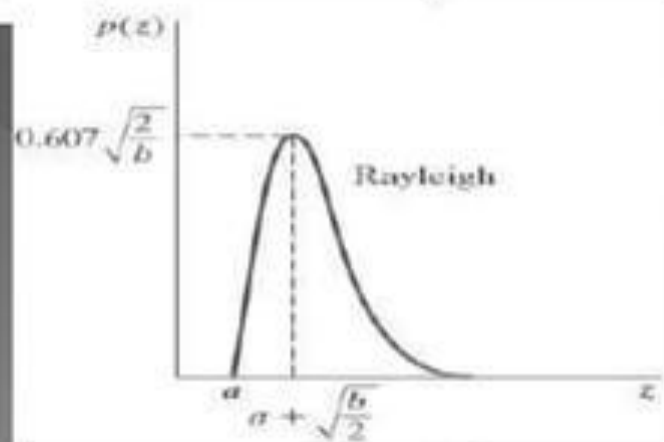
- Rayleigh noise is specified as

$$p(z) = \begin{cases} \frac{2}{b}(z-a)e^{-\frac{(z-a)^2}{b}} & z \geq a \\ 0 & z < a \end{cases}$$

The mean and variance are given by

$$\bar{z} = a + \sqrt{\pi b/4}$$

$$\sigma^2 = \frac{b(4-\pi)}{4}$$



Rayleigh noise

- ⊙ Rayleigh noise PDF is helpful in characterizing noise phenomena in range imaging. (e.g-X-ray,ultraviolet imaging which depend upon the frequency of light)

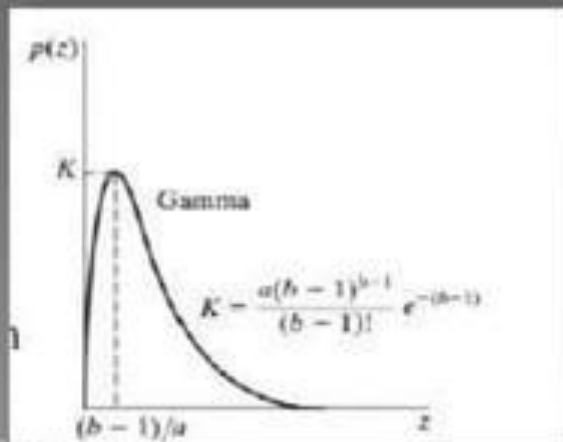
Erlang noise(Gamma noise)

- Erlang noise is specified as

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

Here $a > 0$ and b is a positive integer. The mean and variance are given by

$$\begin{aligned} \bar{z} &= b/a \\ \sigma^2 &= b/a^2 \end{aligned}$$



Erlang noise(Gamma noise)

- Gamma noise finds in laser imaging.

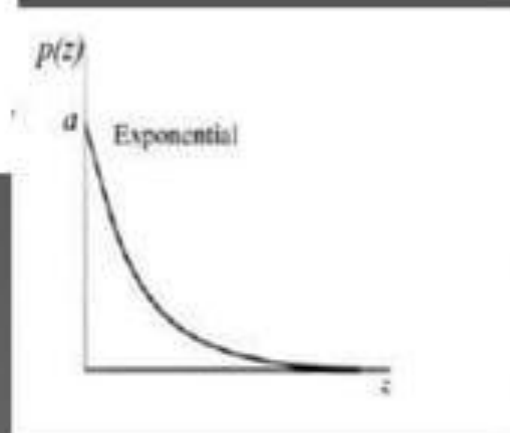
Exponential noise

Exponential noise is specified as

$$p(z) = \begin{cases} ae^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

Here $a > 0$. The mean and variance are given by

$$\begin{aligned} \bar{z} &= 1/a \\ \sigma^2 &= 1/a^2 \end{aligned}$$

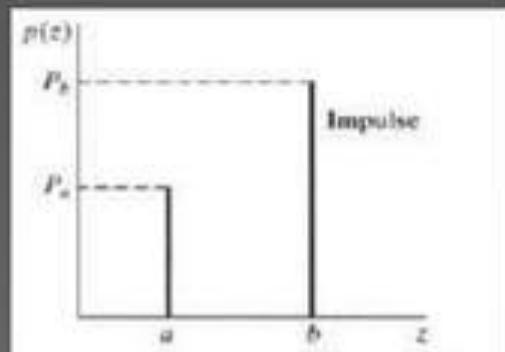


Exponential pdf is a special case of Erlang pdf with $b=1$. Used in laser imaging.

Impulse (salt-and-pepper)

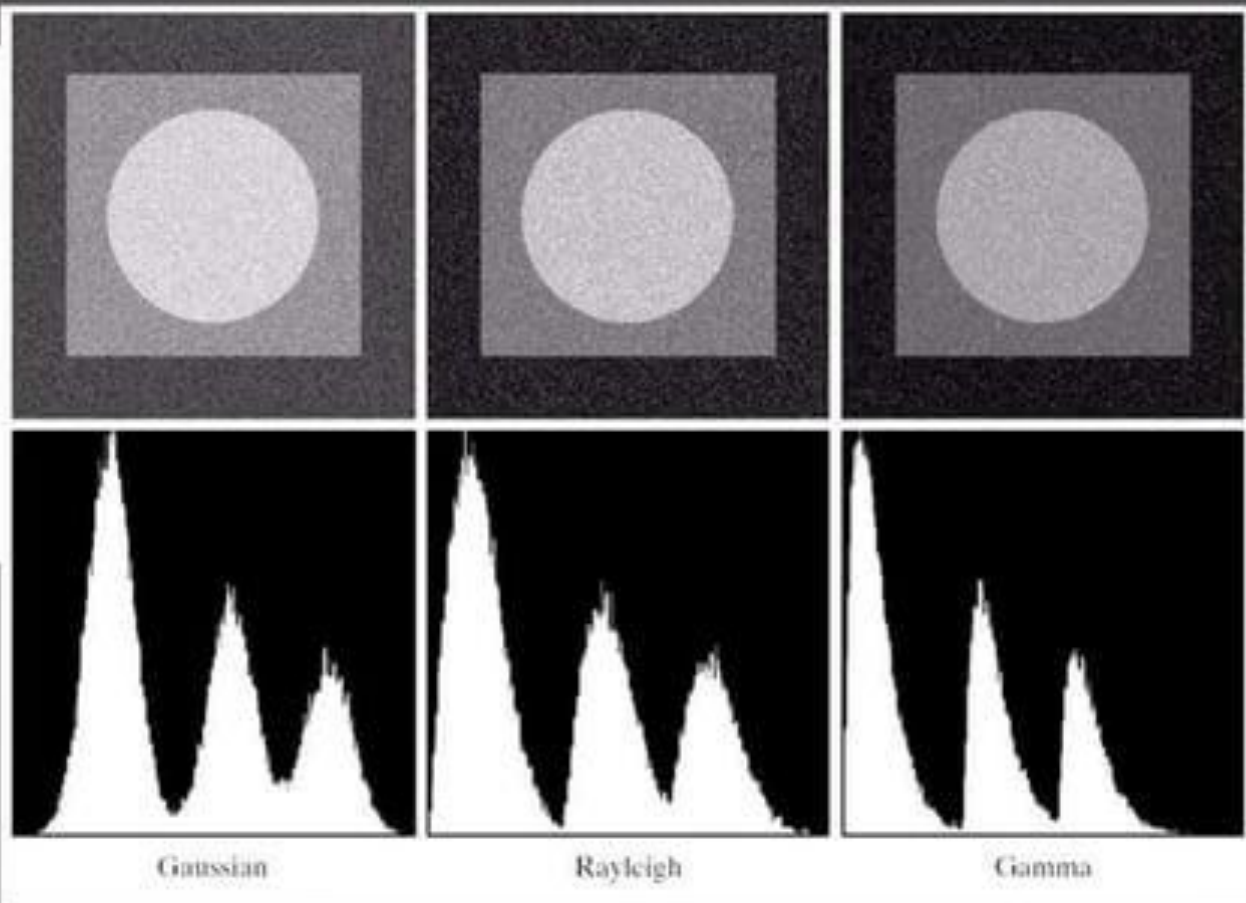
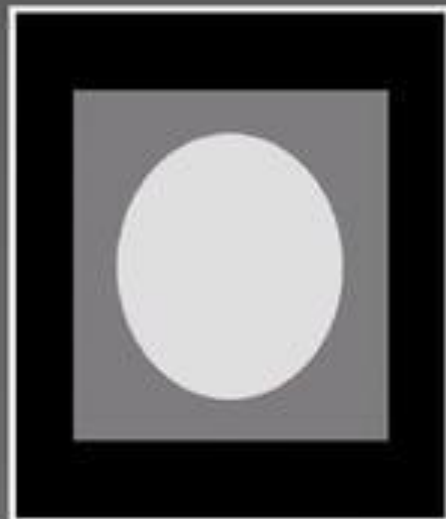
- **Impulse (salt-and-pepper) noise (bipolar)** is specified as

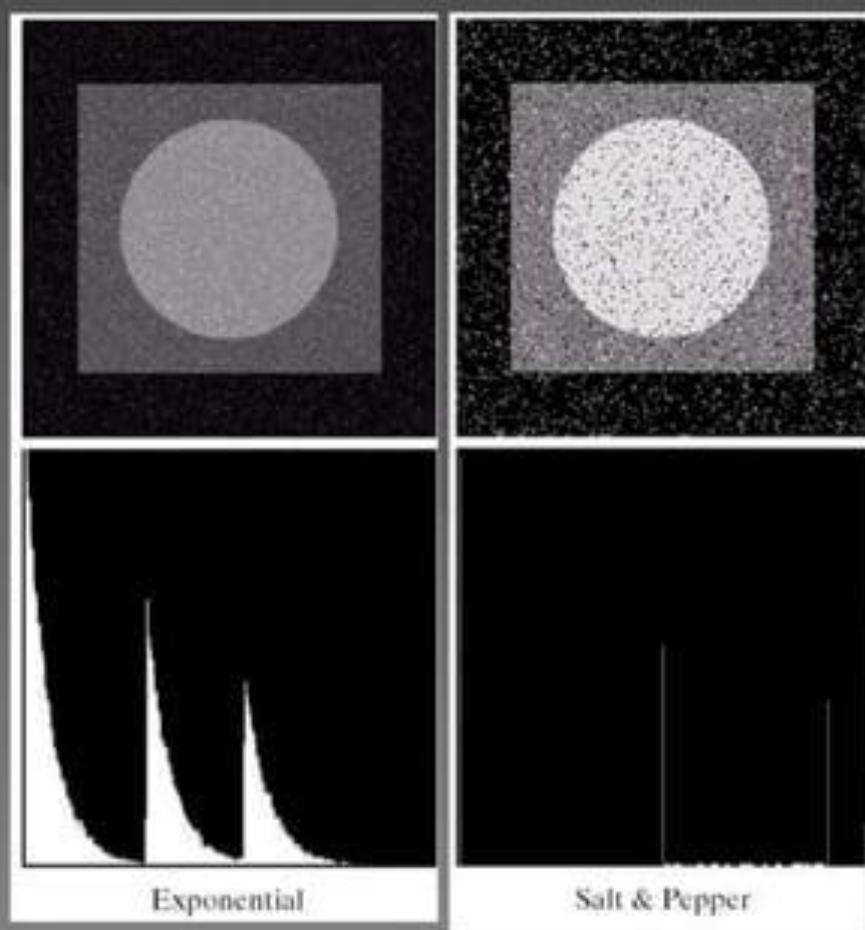
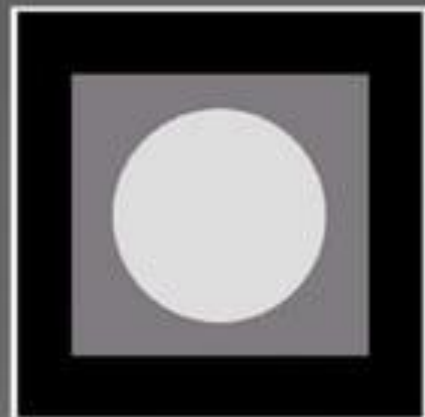
$$p(z) = \begin{cases} P_a & z = a \\ P_b & z = b \\ 0 & \text{otherwise} \end{cases}$$



If $b > a$, intensity b will appear as a light dot on the image and a appears as a dark dot. If either P_a or P_b is zero, the noise is called *unipolar*. If neither probability is zero, and especially if they are approximately equal, impulse noise value will resemble salt and pepper granules randomly distributed over the image. For this reason, bipolar impulse noise is called salt and pepper noise.

Noise models – examples





Noise Removal Restoration Method in spatial domain

- Mean filters

- Arithmetic mean filter
- Geometric mean filter
- Harmonic mean filter
- Contra-harmonic mean filter

- Order statistics filters

- Median filter
- Max and min filters
- Mid-point filter
- alpha-trimmed filters

- Adaptive filters

- Adaptive local noise reduction filter
- Adaptive median filter

Mean filters

⊙ Arithmetic mean filter:

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s, t)$$

The arithmetic mean filter computed the average value of the corrupted image $g(x,y)$ in the area defined by S_{xy} . Let S_{xy} represent the set of coordinates in a rectangular neighborhood of size $m \times n$, centered at the point (x,y) .

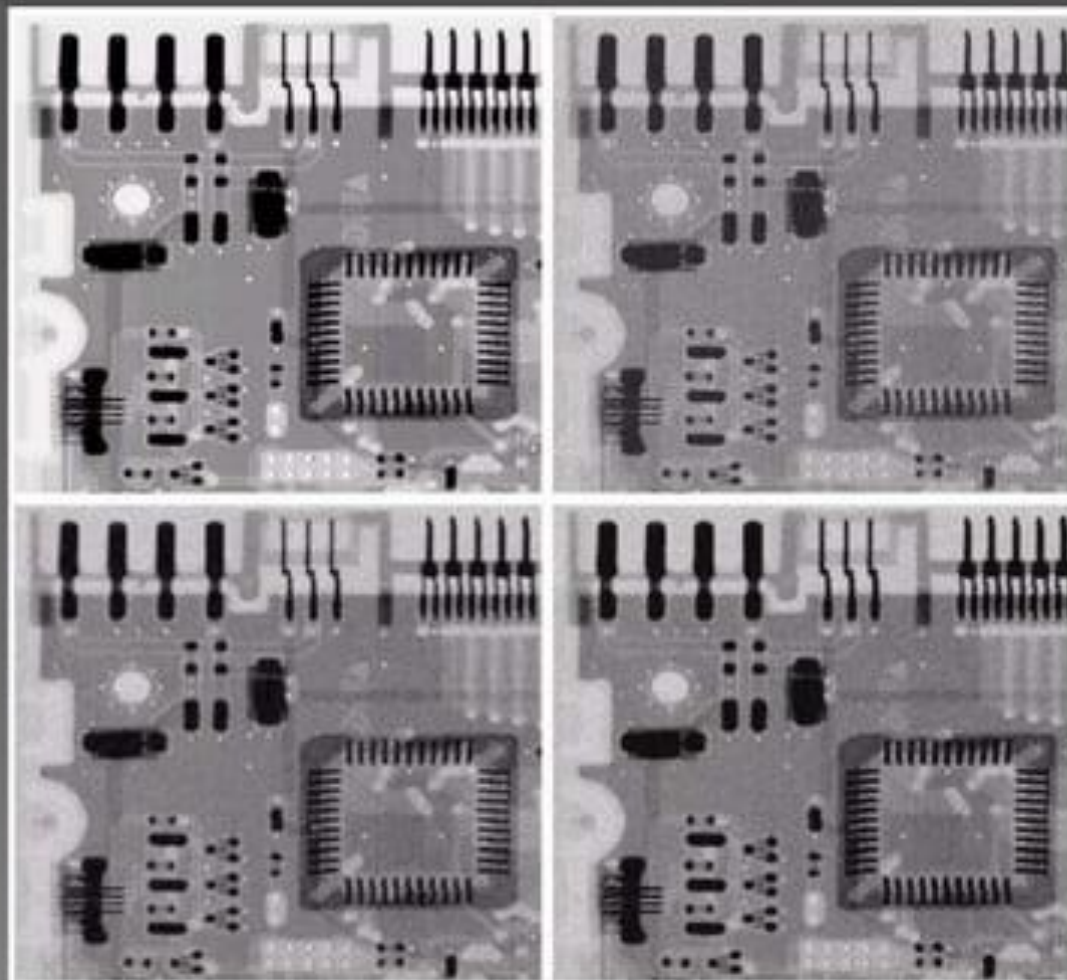
Effect: The Mean filter simply smoothes the variations in an image. noise is reduced as a result of blurring

Geometric mean filter:

$$\hat{f}(x, y) = \left[\prod_{(s,t) \in S_{xy}} g(s, t) \right]^{\frac{1}{mn}}$$

Effect:

Geometric mean filter achieves smoothing comparable to the arithmetic mean filter but it preserves more details (It means loss less image detail in the process)



a b
c d

FIGURE 5.7 (a) X-ray image. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3×3 . (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Harmonic mean filter:

$$\hat{f}(x, y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s, t)}}$$

Effect:

Harmonic mean filter works well for salt noise and other types of noise (such as Gaussian) but fails for pepper noise.

Contraharmonic mean filter:

$$\hat{f}(x, y) = \frac{\sum_{(s,t) \in S_{xy}} g(s, t)^{Q+1}}{\sum_{(s,t) \in S_{xy}} g(s, t)^Q}$$

Here Q is the **order** of the filter. This filter is well suited for reducing the effects of salt-pepper noise. For positive values of Q , eliminates pepper noise; for negative values of Q , it eliminates salt noise. This filter cannot reduce both simultaneously.

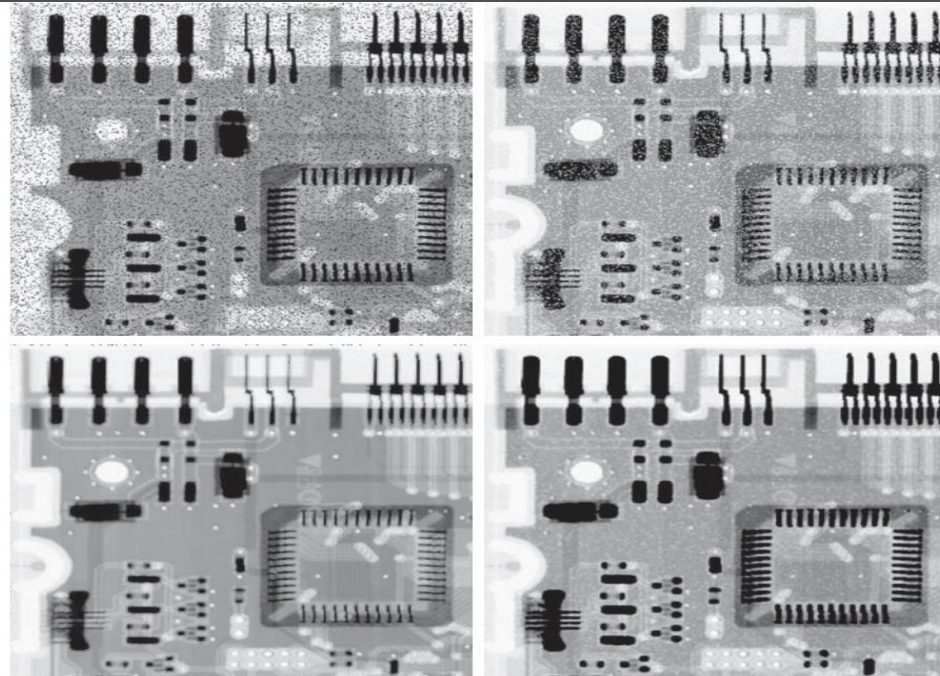
Notice that contraharmonic filter reduces to the arithmetic mean filter when $Q = 0$ and to the harmonic mean filter if $Q = -1$.

Contra-harmonic mean filter:

a b
c d

FIGURE 5.8

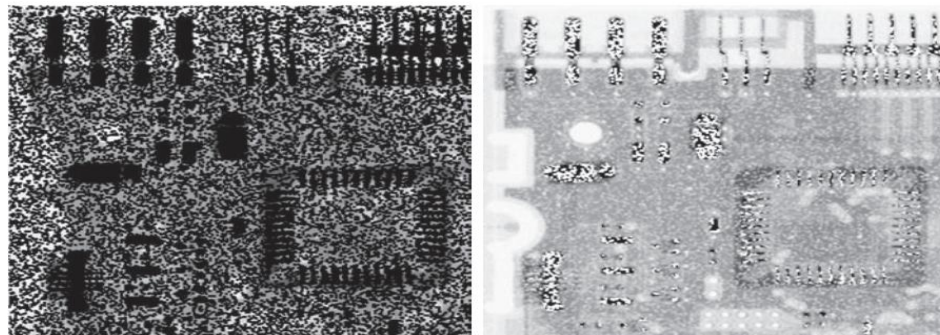
(a) Image corrupted by pepper noise with a probability of 0.1. (b) Image corrupted by salt noise with the same probability. (c) Result of filtering (a) with a 3×3 contra-harmonic filter $Q = 1.5$. (d) Result of filtering (b) with $Q = -1.5$.



a b

FIGURE 5.9

Results of selecting the wrong sign in contra-harmonic filtering. (a) Result of filtering Fig. 5.8(a) with a contra-harmonic filter of size 3×3 and $Q = -1.5$. (b) Result of filtering Fig. 5.8(b) using $Q = 1.5$.



Order-statistic filters

- **Median filter:** It replaces the pixel value by the median of the intensity levels in the neighborhood of that pixel:

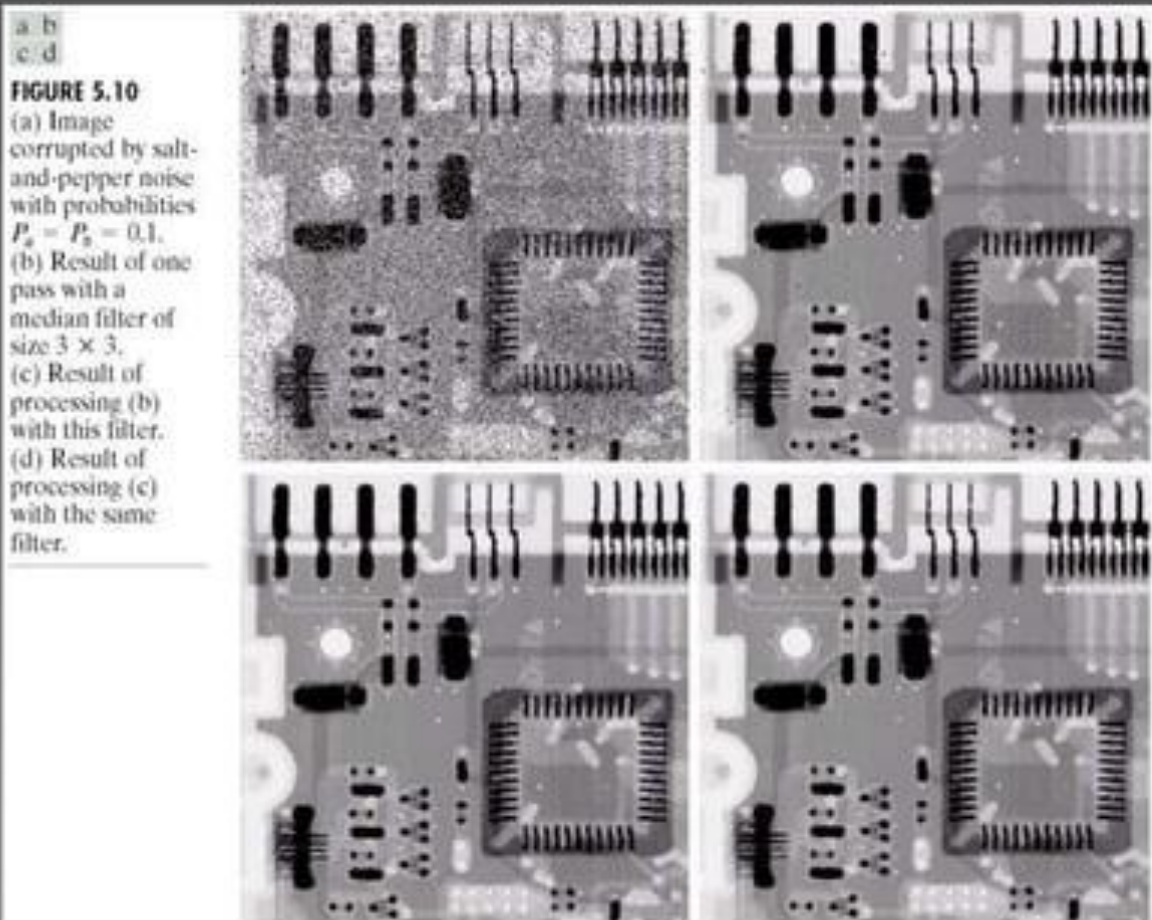
$$\hat{f}(x, y) = \underset{(s, t) \in S_{xy}}{\text{median}} \{g(s, t)\}$$

- **Effect:**
- Median filters provide excellent results for certain types of noise with considerably less blurring than linear smoothing filters of the same size. These filters are very effective against both bipolar and unipolar noise. The same filter can be
- applied more than once to yield better results.

Median Filter

Effective for removing salt-and-pepper noise (impulsive) noise.

$$\hat{f}(x,y) = \underset{(s,t) \in S_{xy}}{\text{median}} \{g(s,t)\}$$



Max and Min Filters

● Max filters:

$$\hat{f}(x, y) = \max_{(s,t) \in S_{xy}} \{g(s, t)\}$$

This filter is useful for finding the brightest points in an image; therefore, its effective against pepper noise.

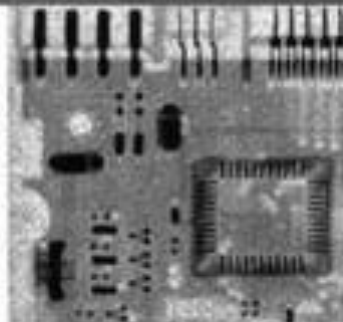
Min filters:

$$\hat{f}(x, y) = \min_{(s,t) \in S_{xy}} \{g(s, t)\}$$

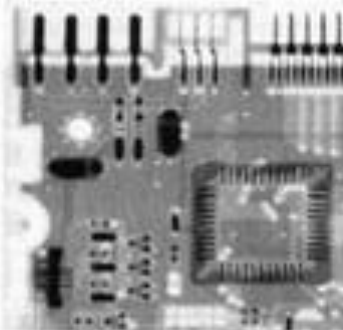
This filter is useful for finding the darkest points in an image; therefore, its effective against salt noise. (it reduces the salt noise because it will eliminate the higher gray values in the subimage.

Max and Min Filters

Image
corrupted
by pepper
noise with
a 0.1
probability

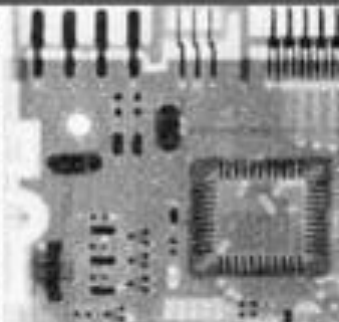


Result of
filtering
with a 3x3
max filter

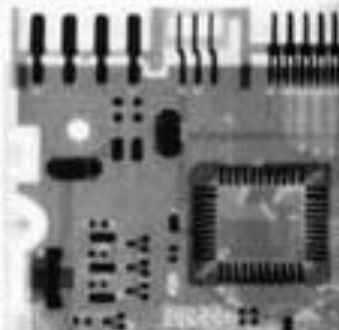


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Image
corrupted
by salt
noise with
a 0.1
probability



Result of
filtering
with a 3x3
min filter



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Midpoint filter:

- It computes the midpoint between the maximum and minimum values of intensities:

$$\hat{f}(x, y) = \frac{1}{2} \left[\max_{(s,t) \in S_{xy}} \{g(s,t)\} + \min_{(s,t) \in S_{xy}} \{g(s,t)\} \right]$$

This filter is a combination of order statistics and averaging and works best for Gaussian and uniform noise contaminations.

Alpha-trimmed mean filter:

- if we delete $d/2$ highest intensity values and $d/2$ lowest intensity values of $g(s,t)$ in the neighbourhood s_{xy} , denote the rest as $g_r(s,t)$, a filter that averages what is left is alpha-trimmed mean filter:

$$\hat{f}(x,y) = \frac{1}{mn-d} \sum_{(s,t) \in S_{xy}} g_r(s,t)$$

d can range from 0 to $mn-1$. When $d = 0$, this filter reduces to the arithmetic mean filter, when $d = mn-1$, this filter reduces to a median filter. For other values of d , the filter is useful in situation with noise of multiple types, such as a combination of salt-and-pepper and Gaussian noise.

Image
corrupted by
additive
uniform noise

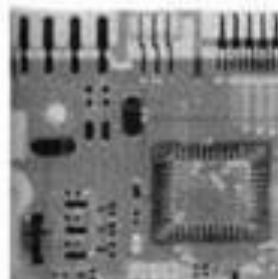
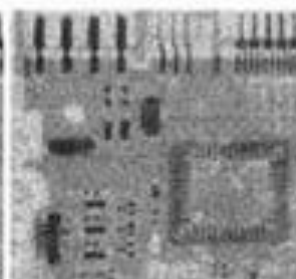
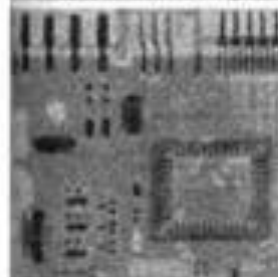


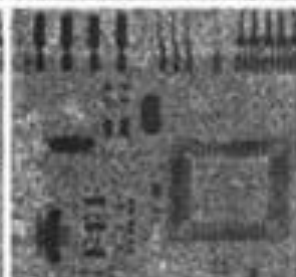
Image
additionally
corrupted by
salt-and-
pepper noise



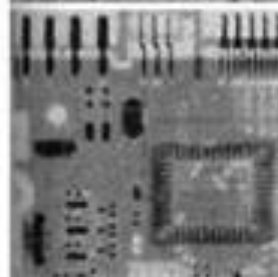
Filtered by a
5x5 arithmetic
mean filter



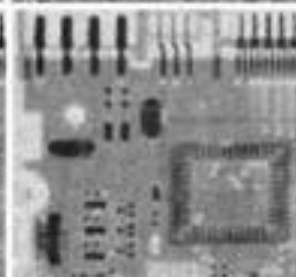
Filtered by a
5x5 geometric
mean filter



Filtered by a
5x5 median
filter



Filtered by a
5x5 alpha-
trimmed mean
filter with $d = 5$



Adaptive filters

- ⊙ Adaptive filters are those filters whose behavior changes based on the statistical characteristics of the image inside the filter region defined by a rectangular window size S_{xy} .
- ⊙ It is better than the mean filter and order statistics filter.
- ⊙ **Two types of filter**
 - Adaptive local noise reduction filter
 - Adaptive median filter

Adaptive local noise reduction filter

- It uses two statistical parameters, mean and variance for the elimination of noise.
- **Mean Parameter:**It gives the average gray value.
- **Variance:**It provides the estimate of the contrast in the image.
- **the adaptive filter is:**

$$\hat{f}(x, y) = g(s, t) - \frac{\sigma_{\eta}^2}{\sigma_L^2} [g(s, t) - m_L]$$

- The response of filter is based on four quantities:-
- $G(x,y)$ the value of noisy image at (x,y) .
- σ_η^2 the variance of noise corrupting $f(x,y)$ to form $g(x,y)$.
- m_L local mean of pixels in the S_{xy}
- σ_L^2 The local variance of the pixels in S_{xy}

$$\hat{f}(x, y) = g(s, t) - \frac{\sigma_\eta^2}{\sigma_L^2} [g(s, t) - m_L]$$

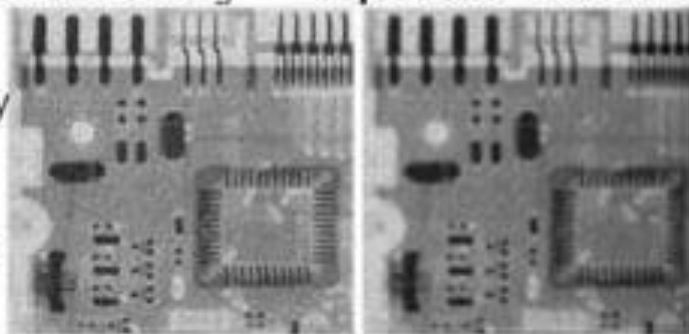
• The behavior of the **Adaptive filter** is obtained as:

-
1. If σ_{η}^2 is zero, the filter returns the value of $g(x,y)$;
 2. If $\sigma_L^2 > \sigma_{\eta}^2$ (typical for edges that needs to be preserved) the filter returns the value close to $g(x,y)$;
 3. If two variances are equal, the filter returns the arithmetic mean value for pixels in the region S_{xy} .

$$\hat{f}(x, y) = g(s, t) - \frac{\sigma_{\eta}^2}{\sigma_L^2} [g(s, t) - m_L]$$

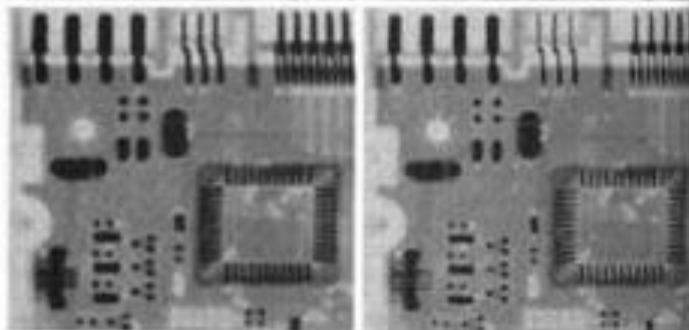
noise only Spatial filtering

Image
corrupted by
Gaussian
noise with
zero mean,
 $\sigma_n^2 = 1000$



Filtered by
a 7x7
arithmetic
mean filter

Filtered by
a 7x7
geometric
mean filter



Filtered by
a 7x7
adaptive
noise
reduction
filter with
 $\sigma_n^2 = 1000$

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Effect

- Adaptive filter achieves approximately the same performance in noise cancellation but adds much less blurring than the mean filters.
- Adaptive filtering typically yields considerably better results in overall performance at the price of filter complexity.

Adaptive median filter

- It can handle impulse noise with larger probabilities than traditional median filter. It operates on a rectangular region S_{xy} , whose *size is changing*. Window size is variable to improve efficiency
- Adaptive median filter has 3 goals:
 - to remove impulse noise,
 - To provide smoothing
 - to reduce distortion

Remember that filtering looks at each original pixel image in turn and generates a new filtered pixel

First examine the following notation:

- Z_{min} = minimum grey level in S_{xy}
- Z_{max} = maximum grey level in S_{xy}
- Z_{med} = median of grey levels in S_{xy}
- Z_{xy} = grey level at coordinates (x, y)
- S_{max} = maximum allowed size of S_{xy}

→ Algorithm works in two levels.

Level A: $A1 = Z_{med} - Z_{min}$
 $A2 = Z_{med} - Z_{max}$.

If $A1 > 0$ AND $A2 < 0$, Go to Level B

Else Increase the window size.

if $window \leq S_{max}$ Repeat Level A
else o/p Z_{xy} .

Level B: $B1 = Z_{xy} - Z_{min}$
 $B2 = Z_{xy} - Z_{max}$

If $B1 > 0$ AND $B2 < 0$; o/p Z_{xy} .

else o/p Z_{med} . → Remove salt & pepper noise

→ Provide Smoothing.

Implementation

- ⊙ Three cases were implemented:
- ⊙ With Salt and Pepper noise alone
- ⊙ With non impulsive noise alone
- ⊙ With both included
- ⊙ Variations in the window size were introduced

Results

Salt and Pepper



Standard Median output



Adaptive median Output

Modified by Yu Hen Hu
Yu Hen Hu

With Non-impulsive noise



Standard Median output



Adaptive Median Output

Modified from <http://www.cse.cuhk.edu.hk/~yuhu/>
Yu Hen Hu

With both types of noise



Gaussian and impulsive Noise



Standard Median output



Adaptive Median output

Modified from restoration.sp4 by
Yu Hen Hu

Effect:

- ⊙ **The adaptive median filter successfully removes impulsive noise from images. It does a reasonably good job of smoothening images that contain non-impulsive noise.**
- ⊙ **When both types of noise are present, the algorithm is not as successful in removing impulsive noise and its performance deteriorates.**

Periodic noise

- This noise typically comes from electrical or electromechanical interference during image acquisition.
- It can be reduced via frequency domain filtering.
- The image is corrupted by sinusoidal noise of various frequencies.
- The parameters of periodic noise are estimated by inspection of Fourier spectrum of the image.



Modified from restoration.ppt by
Yu Hen Hu